Rahul Yadav *Master's in AI Engineering of Autonomous Systems*

yrahuly2000@gmail.com   +49 15510310925   Wolfsburg, Germany
linkedin.com/in/rahul-yadav-14a4ab1b1   06.01.2000

Profile

Master's student in AI Engineering of Autonomous Systems with hands-on experience developing machine learning solutions through academic projects, research, and industry experience. Skilled in machine learning, deep learning, time-series modeling, computer vision, and data analysis, with practical experience in model development, evaluation, and deployment. Experienced in applying Python, PyTorch, TensorFlow, Scikit-learn, and data processing techniques to automotive and autonomous driving applications, including vehicle dynamics and physics-informed machine learning. Seeking full-time opportunities in Machine Learning Engineering, Applied AI, and Data Science.

Education

Master of Engineering, *Technische Hochschule Ingolstadt* — 03/2024 – 08/2026 | Ingolstadt, Germany
Artificial Intelligence Engineering of Autonomous Systems, Grade- 1.9 (awaiting)

Bachelors of Technology, *K J Somaiya College of Engineering (Mumbai University)* — 08/2017 – 07/2021 | Mumbai, India
Mechanical Engineering, Grade- 1.6

Master Thesis

Development of an Data-Driven Vehicle Dynamics Model for XiL Testing Environments — 12/2025 – Present | Wolfsburg, Germany
Volkswagen ADMT, MOIA GmbH
- Developed a LSTM-based vehicle model to estimate vehicle dynamics states from CAN signals.
- Implemented physics-based loss constraints for realistic motion behavior.
- Wrapping this trained network to autoregressively predict vehicle states given the control input, Trained Network -> Vehicle Model.
- Vehicle model achieved 0.90+ R2 score for all signals in open loop evaluation using real world data from proving ground.
- Integrated the vehicle model with Volkswagen's inhouse simulator to run scenarios in closed loop setup with the Self Driving Stack (SDS) in the loop.

Professional Experience

Working Student- SimulationX, *Keysight Inc* — 09/2024 – present | Dresden, Germany
- Developed and enhanced vehicle dynamics models with component-level fidelity.
- Improved energy consumption simulations by modeling tire power loss and suspension behavior.
- Contributed to OpenCRG integration for realistic road surface modeling.

Junior Engineer, *Toyo Engineering India* — 08/2021 – 04/2024 | Mumbai, India
- Designed 3D piping layouts and performed isometric analysis using Aveva E3D for IOCL and Nayara Energy projects.

Projects

Vehicle State Estimation for Autonomous Driving — 03/2025 – 06/2025
- Developed a vehicle state estimation framework using OBD and ADMA sensor data, sampled at different frequencies..
- Preprocessed and aligned asynchronous sensor data to improve model input consistency and training.
- Implemented and compared RNN, LSTM, and Transformer-based architectures for time-series modeling of vehicle states such as velocity, acceleration, and yaw rate.
- Integrated physics-based constraints into ML models to guide learning and improve generalization across diverse driving scenarios.

Robust Visual Perception Pipeline for Autonomous Systems (Using KITTI Dataset) — 08/2026 – Present
Tech: Python, PyTorch, NumPy, Matplotlib, Jupyter, KITTI, CNNs, U-Net, BEV perception, 3D geometry, sensor calibration.
- Converted raw point clouds into bird's-eye-view representations.
- Designed a lightweight U-Net-style encoder-decoder with multi-scale features and CenterNet-inspired heads to predict vehicle centers, dimensions, and orientations.
- Built robust data loading, augmentation, training, decoding, visualization, and full-dataset evaluation pipelines.
- The model achieved 83.3% precision, 89.0% recall, and an 86.0% F1 score on a held-out validation set, with a 0.14 m mean center-localization error and approximately 27 FPS end-to-end throughput.
- Identified sparse long-range LiDAR observations as the primary limitation, motivating ongoing camera–LiDAR sensor-fusion development.

LLM Based Scenario Analysis Tool — 04/2026 – 05/2026
Tech: Claude Code, Claude LLMs, AWS Boto3
- Built an LLM-powered automated analysis tool for 50+ simulation scenarios, replacing a time-intensive manual analysis process.
- Used Claude to evaluate simulation outputs against expected ("perfect") scenario behavior, test context, and known failure conditions to determine pass/fail status and their reasoning.
- Automated generation of a consolidated test-run report, providing scenario-level results and explanations and enabling significantly faster analysis of large simulation runs.

Traffic Sign Recognition (CNN + YOLOv8, GTSRB Dataset) 08/2025

- Achieved 99%+ accuracy on GTSRB dataset using a custom CNN in PyTorch.
- Combined YOLOv8 detection with CNN classification for modular real-time recognition.
- Deployed as a Dockerized FastAPI service with live video stream inference.

Q Learning and DQN 05/2024 – 07/2024

- Developed a custom environment using OpenAI- Gymnasium, the environment has an agent trying to reach goal by evading different hell states.
- Used Q-learning and DQN Reinforcement learning methods to train the agent to follow most suited optimal path.

Past Experiences

Suspension & Vehicle Dynamics Lead, *Orion Racing India, Formula Student* 11/2018 – 07/2021 | Mumbai, India

- Designed a Z-type anti-roll bar, improving system adjustability by 22%.
- Developed Yaw Moment Diagrams and MATLAB models for traction & launch control.
- Conducted tire data analysis (TTC) to optimize vehicle performance.

Skills

Programming — Python, C++ | **Machine Learning & AI** — PyTorch, TensorFlow, Scikit-learn, OpenCV- Computer Vision, LangChain, Hugging Face | **Data Analysis & Visualization** — Pandas, NumPy, Matplotlib | **MLOps & Deployment** — FastAPI, Docker, MLflow, DVC, Grafana | **Data Engineering** — SQL (MySQL), PySpark, Airflow, dbt | **Cloud & Storage** — AWS S3 | **Version Control & Collaboration** — Git, GitHub | **Simulation & Modeling** — CARLA, MATLAB, Simulink | **AI tools** — Claude Code, Codex | **Agile Framework** — Atlassian Confluence, Jira

Publications

IEEE International Transportation Electrification Conference - India 2021

- Presented paper at conference which discussed methods to implement Launch Control (LC) in an Formula Student Electric Vehicle.

International Journal for Research in Applied Science and Engineering Technology 2022

- Partial Differential Equations (PDE) toolbox from MATLAB was used to perform Finite Element Analysis (FEA) and the results were compared with more common FE software.

Languages

English ●●●●●

German ●●●○○
B1-Level

Hindi ●●●●●